Competitive variant-effect prediction with a 1B genomic language model trained using functional-region mixtures and transferred hyperparameters

TODO

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Abstract

Genomic language models can learn functional constraints directly from DNA sequence, but strong performance has often depended on specialized architectures, very large training budgets, or information derived from alignments and functional genomics. We use MarinDNA, a standard decoder-only Transformer, to study functional-region mixtures, hyperparameter transfer, and parameter scaling. Hyperparameters calibrated on approximately 25M-parameter reference models identified the best observed learning rate at 255M, 476M, and 1B parameters, and eight models from 46M to 4B parameters showed monotonic loss reduction with a Kaplan power-law fit ($R^2 = 0.999$). Continued training on a five-region mixture yielded the 1B m5.1 model after 166B tokens. On the frozen Mendelian variant-effect benchmark, m5.1 reached 39.49% macro AUPRC compared with 38.24% for Evo 2 40B, a paired difference of 1.25 percentage points (95% cluster-bootstrap CI −1.93 to 4.46; two-sided bootstrap $p = 0.440$). m5.1 used approximately 1,980-fold fewer reported training FLOPs and achieved approximately 2,330-fold higher as-deployed scoring throughput than Evo 2 40B at each model's native context length; alignment-based and functionally supervised models remained stronger overall. These comparisons cover human variant-effect prediction with a 255-base model context and do not establish long-range or cross-species performance.

1. Introduction

Genomic language models (gLMs) learn statistical constraints directly from DNA sequence and can support variant-effect prediction, sequence design, and transfer learning (Benegas, Ye, et al., 2025). Early sequence-only models showed that such constraints can predict the effects of genome-wide variants without task-specific labels (Benegas et al., 2023). More recent work has expanded model scale, context length, biological coverage, and architectural specialization, but it remains unclear how much of the resulting performance depends on specialized architectures rather than the data and optimization recipe.

The distinction matters because the strongest genomic predictors often use information that is unavailable outside a small number of well-studied species. GPN-Star explicitly models whole-genome alignments and phylogeny (Ye et al., 2025), whereas AlphaGenome is supervised on thousands of human and mouse functional-genomics tracks (Avsec et al., 2026). Sequence-only models require neither input at inference time and can therefore be applied wherever an assembled genome is available. They do not yet replace alignment-based or functional-genomics-supervised models in humans, but they may provide a practical route to genome-wide constraint prediction in species that lack those resources. Human evaluation is consequently a stringent proxy for this broader goal, not direct evidence of cross-species transfer.

We treat training-data construction as a primary modeling variable. TraitGym found that a 152M-parameter promoter specialist was competitive with Evo 2 40B on promoter variants, while Evo 2 scaling produced smaller gains on distal regulatory variants than on several other consequence classes (Benegas, Eraslan, et al., 2025). Those observations motivate explicit control over functional-region sampling rather than uniform sampling from mostly neutral genomic background. MarinDNA begins with coding, upstream, and downstream regions available from standard gene annotations, then adds non-coding RNA and candidate cis-regulatory elements by projecting human annotations across species. The experiments vary which regions are present, their mixture weights, and the stage of training at which they are introduced.

To isolate those choices, we use a conventional decoder-only Transformer rather than a genomics-specific sequence operator. The model is a nucleotide-level Qwen3 causal language model with a 255-base context, chosen to model individual functional elements while keeping training and variant scoring inexpensive. An initial objective comparison favored causal language modeling for subsequent development, but was not a definitive matched-compute comparison. Long-range regulatory modeling and bidirectional adaptation remain outside the scope of this study.

We evaluate frozen checkpoints on clinically curated Mendelian variants and saturation-genome-editing measurements. Each model is assessed through two complementary readouts: a zero-shot reference-versus-alternate sequence log-likelihood ratio and a chromosome-disjoint linear probe on paired allele embeddings. The likelihood score tests whether the training objective itself ranks deleterious alleles; the probe tests whether variant-relevant information is present in the learned representation even when it is not exposed by sequence likelihood. These evaluations measure human functional constraint and pathogenicity, not gene-expression changes or organism-level fitness.

Evo 2 40B is the principal sequence-only baseline because it is broadly evaluated across genomic regions and was trained at a substantially larger scale: 40B parameters, approximately 9.3T DNA tokens, and 2.25e24 reported training FLOPs (Brixi et al., 2026). Against that baseline, we ask whether a simpler and much smaller model can be made competitive through data curation and transferable optimization. Optimizer settings calibrated at small scale transfer across model size, token horizon, and batch size. Validation loss decreases monotonically across the tested parameter range, while zero-shot Mendelian missense performance peaks at 128M parameters. A staged five-region training mixture yields a 1B-parameter model that is statistically competitive with Evo 2 40B on the frozen Mendelian benchmark, uses 1,982-fold fewer reported training FLOPs, and achieves approximately 2,330-fold higher native-context scoring throughput.

2. Results

2.1. Functional-region sampling

Equal upstream/CDS sampling retained performance across both region types, whereas proportional 10/90 sampling behaved similarly to CDS-only training. Across the study, training examples came from annotation-derived coding, upstream, and downstream regions and from alignment-projected non-coding RNA and candidate cis-regulatory-element annotations. Figure 1 summarizes the provenance and scale of these datasets; individual experiments used the versions and mixtures specified below and in the supplementary provenance table.

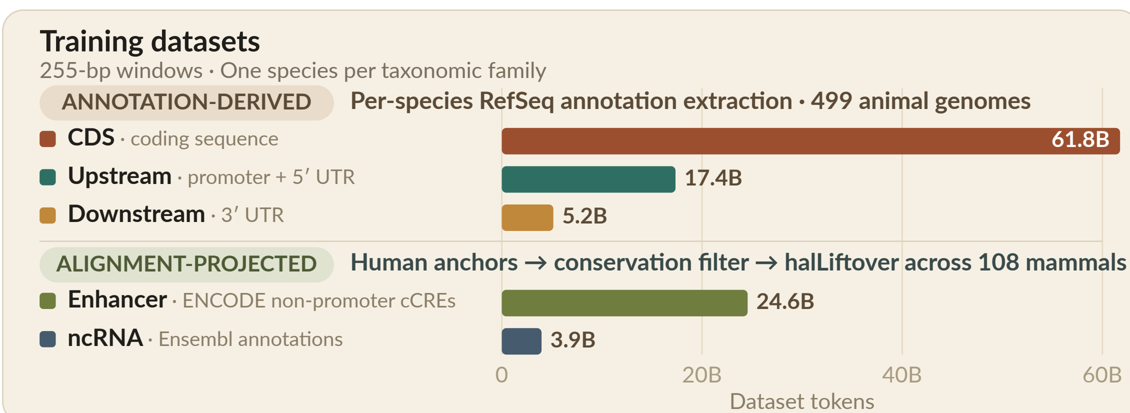

Figure 1 **Training-dataset provenance and scale.**

We first trained a 1.7B upstream-region specialist following the region-specialist premise of GPN-Promoter (experiment #21).[1] The upstream specialist and a corresponding CDS specialist (experiment #27) produced point estimates similar to Evo 2 40B on their matched consequence classes, although GPN-Star remained stronger. Both specialists preceded the systematic hyperparameter-transfer recipe developed below.

[1]The upstream and CDS datasets used in these early experiments were earlier versions of those summarized in Figure 1: broadly comparable, but not identical. These models also used a 512-bp context without a BOS token, roughly twice the 255-bp context adopted for the later recipe.

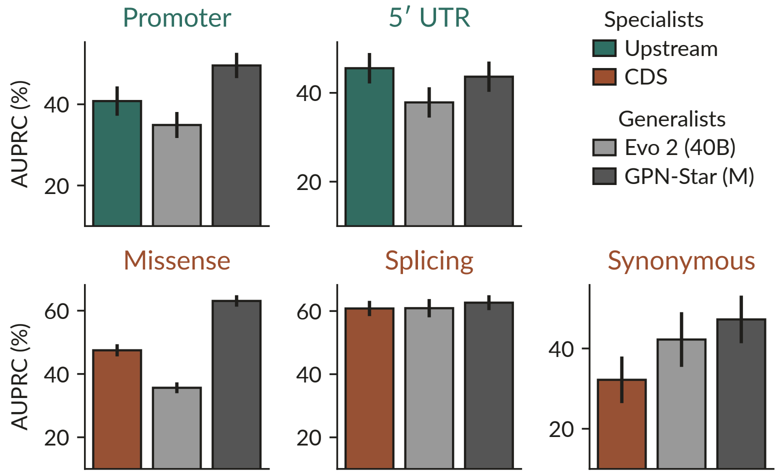

Figure 2 **Region-matched zero-shot Mendelian VEP: specialists versus generalists.** Error bars denote SE.

We next compared proportional and equal sampling in one model (experiment #13). Pooling the datasets without reweighting gave 10% upstream and 90% CDS examples and behaved similarly to CDS-only training. Equal 50/50 sampling retained useful performance on both regions. The experiment did not test enough mixture weights to establish 50/50 as optimal; region datasets differ in both size and density of learnable biological signal.

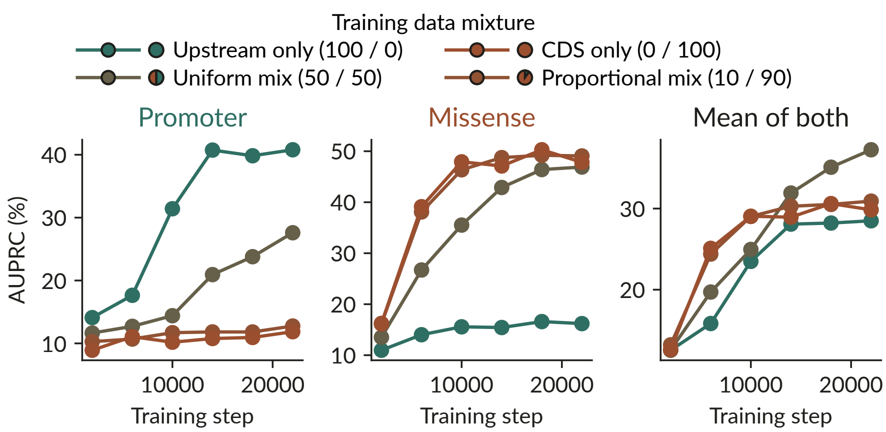

Figure 3 **Upstream/CDS training-mixture comparison.** Zero-shot results; the right panel is the unweighted mean of the promoter and missense AUPRC values.

2.2. Hyperparameter transfer

Our first attempt to scale manually lowered the learning rate as the model grew from 0.6B to 1.7B and 4B parameters (experiment #57). The larger models did not improve over the 0.6B model, and the early 4B run became unstable. That failure made systematic hyperparameter transfer a prerequisite: without a trustworthy optimization recipe, a model-size comparison would confound scale with tuning quality.

The annotation-derived DNA pool available at the time contained ~85B tokens. Figure 4 shows its proportional CDS, upstream, and downstream composition.

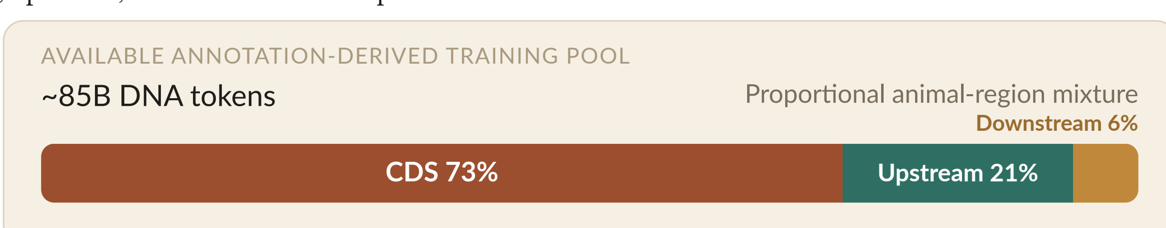

Figure 4 **Available annotation-derived training pool.**

The available pool was small relative to modern accelerator-era training corpora, and we did not have consistent access to preemptible Google TPU Research Cloud slices beyond roughly 32 H100s of peak FLOPs. We therefore targeted O(100B)-token runs that could be reproduced at academic compute scale. This regime permits repeated exposure to the data and may violate compute-constrained transfer assumptions at an unknown rate.

We did not identify an established transfer framework for this data-constrained regime. Following the pattern used in Delphi, we fit a small reference sweep with the Vizier Bayesian optimization framework and scaled the result using a Complete(d)-inspired AdamH heuristic.[2]

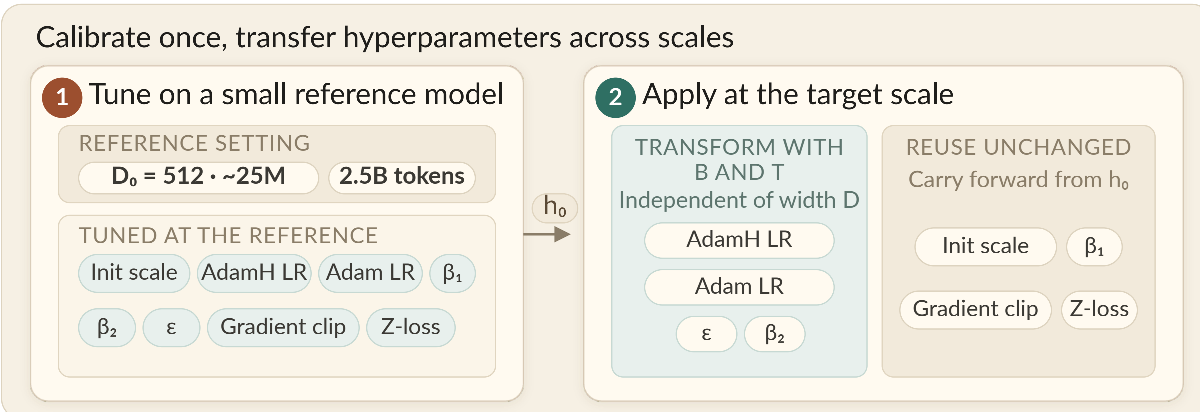

Figure 5 **Hyperparameter calibration and transfer.**

Figure 5 separates reference calibration from target application. A fixed rule maps hidden width D to model geometry, and a second rule maps reference optimizer settings to a new batch size B and token horizon T. The reference sweep tunes initialization scale, the two peak learning rates, β_1, β_2, ε, gradient clipping, and z-loss. At the target, the two learning rates, β_2, and ε are transformed with B and T, while initialization scale, β_1, gradient clipping, and z-loss are reused unchanged. The transfer equations, schedule, and implementation references are given in Section 4.4.

The reference sweep used ~25M-parameter models trained for 2.5B tokens with a 16k-token batch, or roughly 4e17 FLOPs per run. We validated the transferred hyperparameters at 255M, 476M, and 1B parameters using four times as many tokens, one quarter of the batch size, and roughly 170 times the FLOPs per run. The transferred learning rate gave the lowest final validation loss among tested settings at all three target scales (Figure 6); the less sensitive optimizer hyperparameters are shown separately in Figure 7.

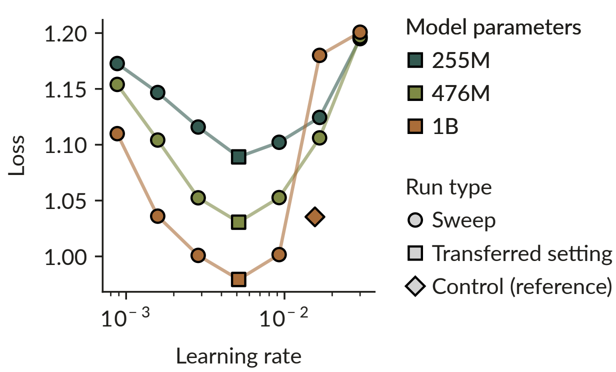

Figure 6 **Learning-rate transfer across model scales.** Results across the 255M, 476M, and 1B validation scales. The control denotes the best configuration found in the smaller reference sweep. At each target scale, the transferred learning rate gives lower final validation loss than the control and the other tested learning rates.

[2]Complete(d) refers to the compute-constrained hyperparameter-transfer framework described in Complete(d): Data-Optimizing Hyperparameter Transfer.

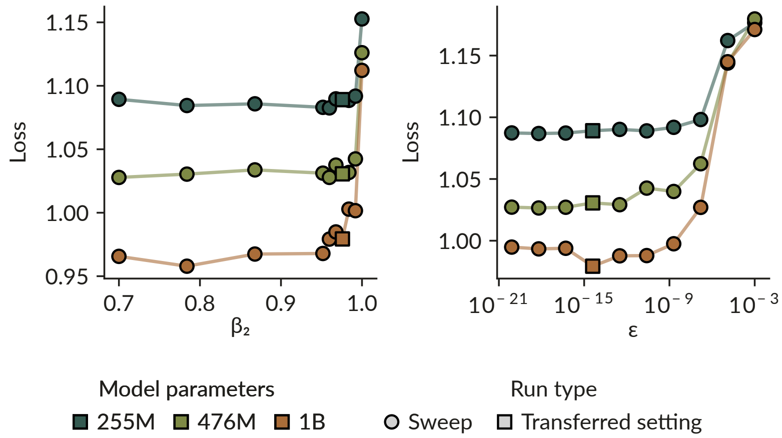

Figure 7 **Adam β_2 and ε transfer.** Results across the same scales as Figure 6.

Prior biology foundation-model work has used μP-style transfer, but we are unaware of a DNA study evaluating optimizer transfer across both token horizon and batch size. Adam β_2 showed a possible overcorrection at 1B parameters; the other transferred settings showed no qualitative failure across the tested scales. Region-stratified results were also qualitatively consistent across CDS, upstream, and downstream sequence (Supplementary Figure S1). These observations support using the transferred recipe for the controlled parameter-scaling comparison. The tested grids do not establish global optimality or validate the heuristic outside the evaluated scales and training regimes.

2.3. Parameter scaling

Training and validation loss decreased monotonically across eight models from 46M to 4B parameters, each trained on ~84B tokens. The final validation losses followed a Kaplan-style power law with $R^2 = 0.999$ over the tested range.[3] The sweep used the same training recipe at every model size, with hyperparameters set by the transfer heuristic above. It consumed ~4.3e21 FLOPs in total; the 4B run consumed ~2.1e21 FLOPs and took approximately three weeks.[4] The validation statistic and its limitations are defined in Section 4.5.

[3] This follows the empirical scaling-law setup from Kaplan et al., where model loss is fit as a predictable function of model size, data, and compute.

[4] The 4B run matches the compute used at the same model scale in the data-constrained scaling study by Muennighoff et al. (NeurIPS 2023).

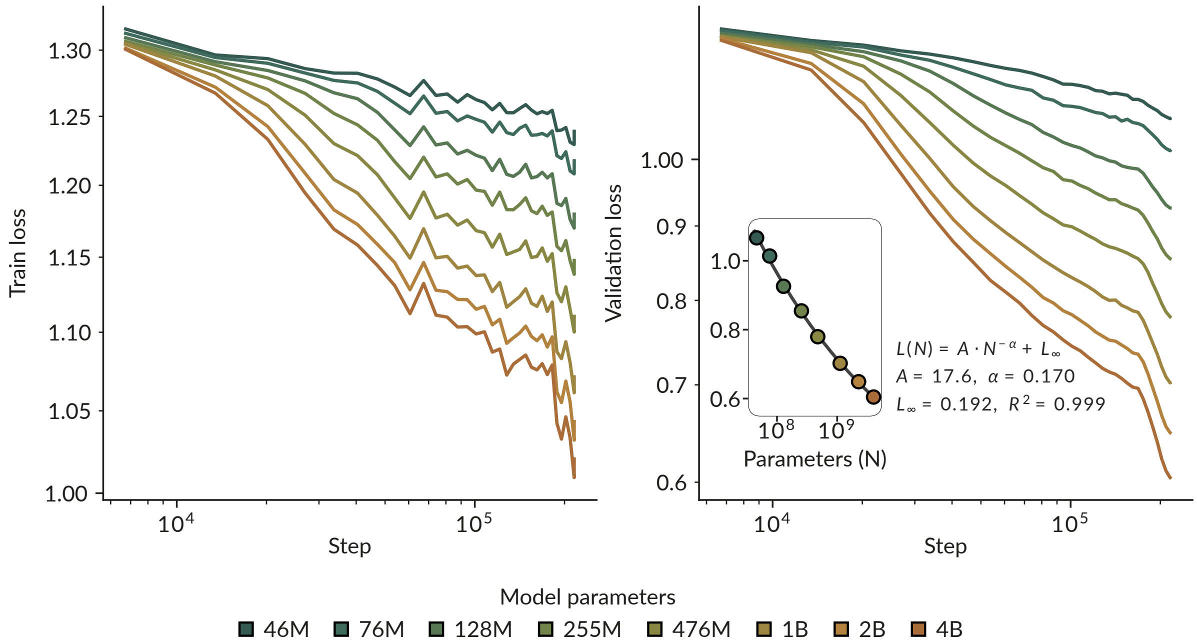

Figure 8 **Loss scaling across model sizes.** Training and validation loss for eight models from 46M to 4B parameters, each trained on ~84B tokens; the inset shows a Kaplan power-law fit to final validation loss.

Training remained stable at every scale (Figure 8). The WSD learning-rate schedule uses 10% warmup and 20% decay, producing the visible loss drop over the final 20% of tokens. The scaling fit describes the tested parameter range; downstream evaluation tests whether lower validation loss corresponds to better VEP performance.

2.4. Downstream performance

When zero-shot LLR and linear probes are evaluated on identical variants, performance improves with scale for most variant types. Mendelian missense is the clearest exception: zero-shot LLR peaks at 128M parameters and then deteriorates, even as linear-probe performance continues to improve.[5] SGE missense improves with scale under both scoring protocols, so the regression is specific to zero-shot Mendelian missense in these experiments.

[5] Non-monotonic likelihood-based zero-shot variant-effect performance with increasing model scale has previously been observed in other settings. Gordon et al. report that ESM-2 performance on protein deep-mutational-scanning benchmarks degrades beyond an intermediate model size and show that performance depends on the likelihood assigned to the wild-type sequence. Pugh et al. similarly report plateaus or regressions for larger protein language models under standard likelihood-based scoring. These studies concern protein models and different evaluation regimes; we do not know whether our Mendelian missense result has the same cause.

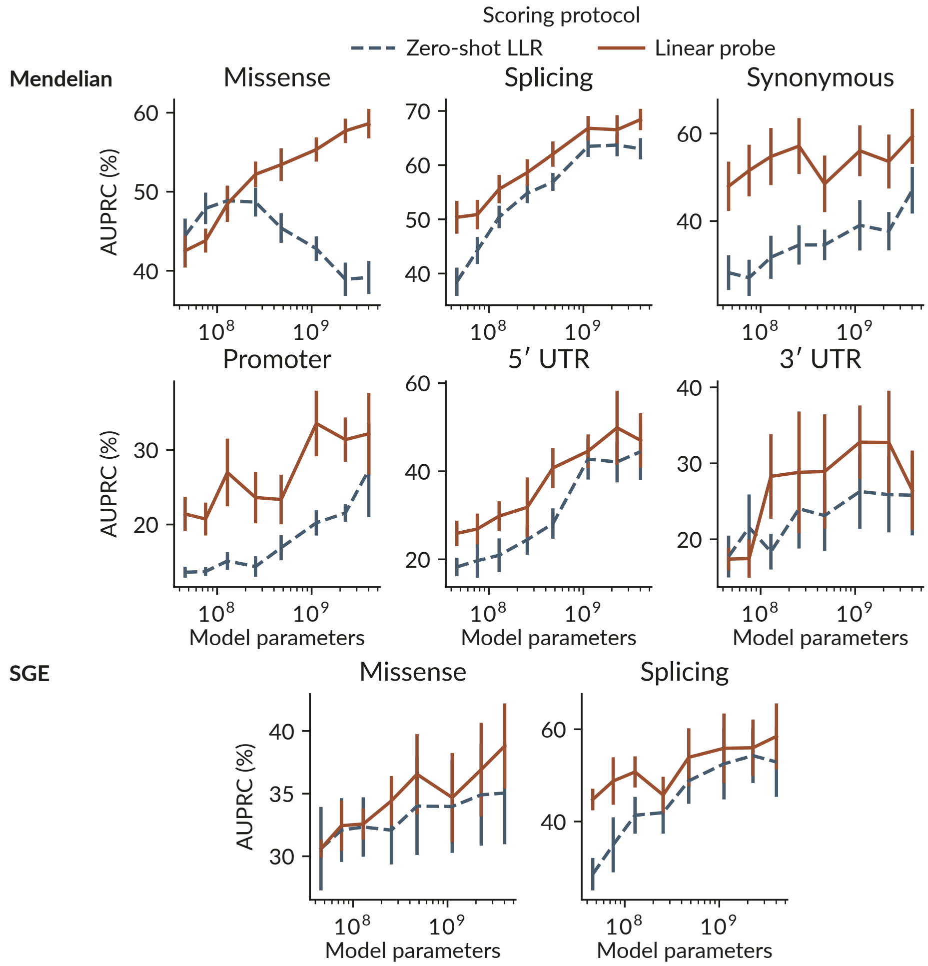

Figure 9 **VEP performance across model scale.** Zero-shot LLR and a linear probe are compared on identical variants. Performance is measured as chromosome-weighted AUPRC; error bars denote SE.

Plotting the same results against matched-region validation log-likelihood gives the same picture. For all other combinations of variant type and scoring protocol, better validation log-likelihood is associated with better downstream performance. Zero-shot Mendelian missense again points in the opposite direction: performance declines even as validation log-likelihood improves.

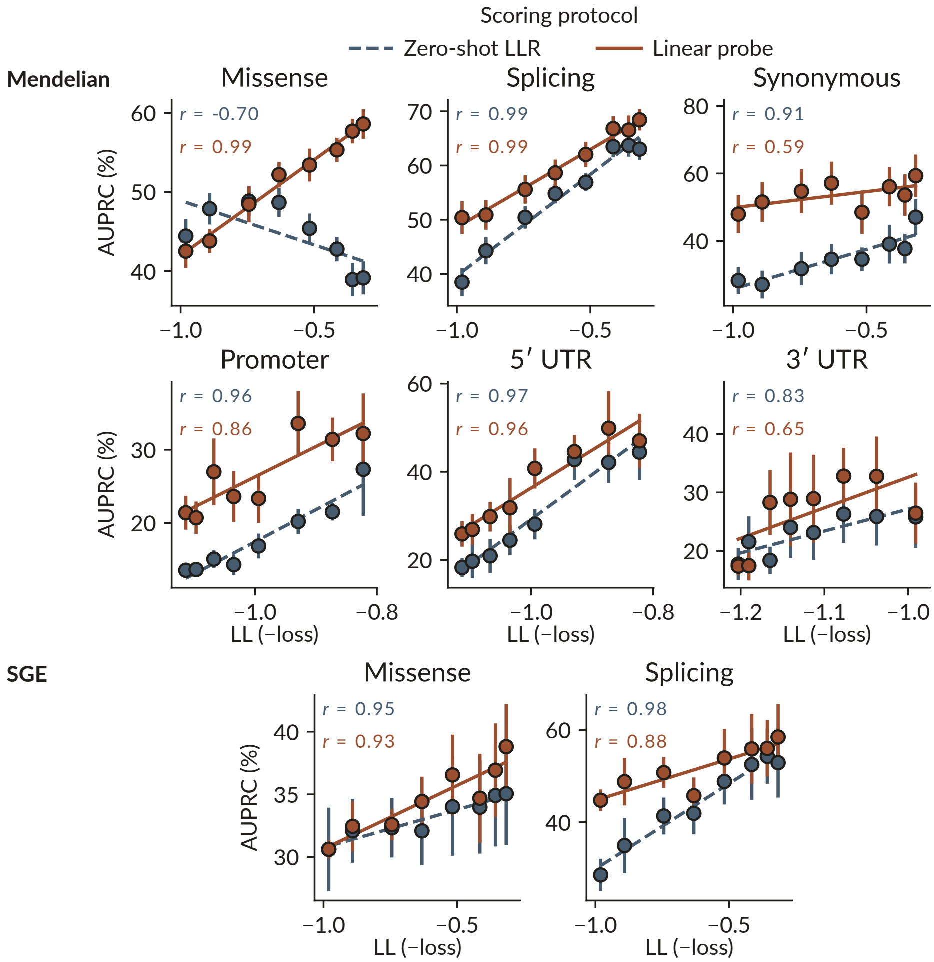

Figure 10 **VEP performance versus validation log-likelihood.** Matched-region LL (shown as −loss) across the eight parameter-scaling endpoints. Performance is measured as chromosome-weighted AUPRC. Lines are least-squares fits, and r denotes Pearson correlation; error bars denote SE.

Evo 2 shows the same readout divergence on the Mendelian missense benchmark: linear-probe performance improves with model size while zero-shot LLR performance declines. The recurrence is confined to Mendelian missense in these two model families and does not show that zero-shot readouts generally deteriorate with scale. Zero-shot LLR alone can therefore miss improvements present in frozen representations for this task.

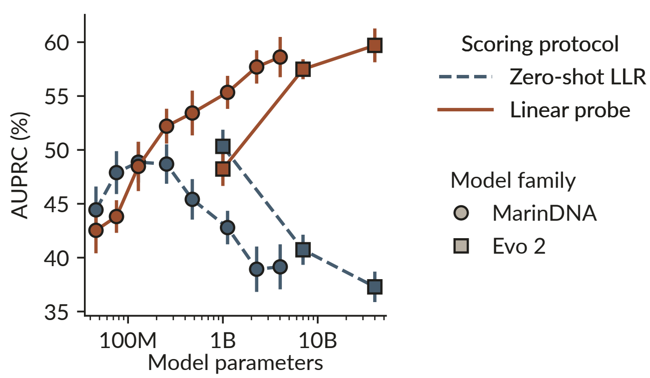

Figure 11 **Missense readouts across model scale.** MarinDNA and Evo 2 are compared using zero-shot LLR and a linear probe on identical Mendelian variants. Performance is measured as chromosome-weighted AUPRC; error bars denote SE.

2.5. Five-region mixture lineages

The later mixture experiments retained the transferred optimizer settings but were exploratory. We changed mixture constituents during training, allowed repeated exposure to the data, and selected continuations in response to observed performance gaps. The resulting lineages do not constitute a controlled curriculum ablation. We standardized these experiments on a 1B-parameter model.[6]

Our starting point was a uniform mixture of CDS, upstream, and downstream sequence. These were the three region datasets we could initially construct consistently across many species from standard genome annotations. The scaling study had instead sampled these regions in proportion to dataset size. Echoing the earlier mixture experiments, that recipe produced good CDS performance but left a large gap on the less abundant upstream tasks. We therefore switched to uniform weighting so that each functional region received meaningful exposure.

We compare a staged lineage (m5.1) with two controls trained on five-region mixtures from the beginning (m1.3 and m3.3). m5.1 first trained for ~104B tokens on the uniform three-region mixture. After alignment projection made multi-species ncRNA and enhancer datasets available, we added them as two equally weighted regions and continued training for another ~62B tokens.

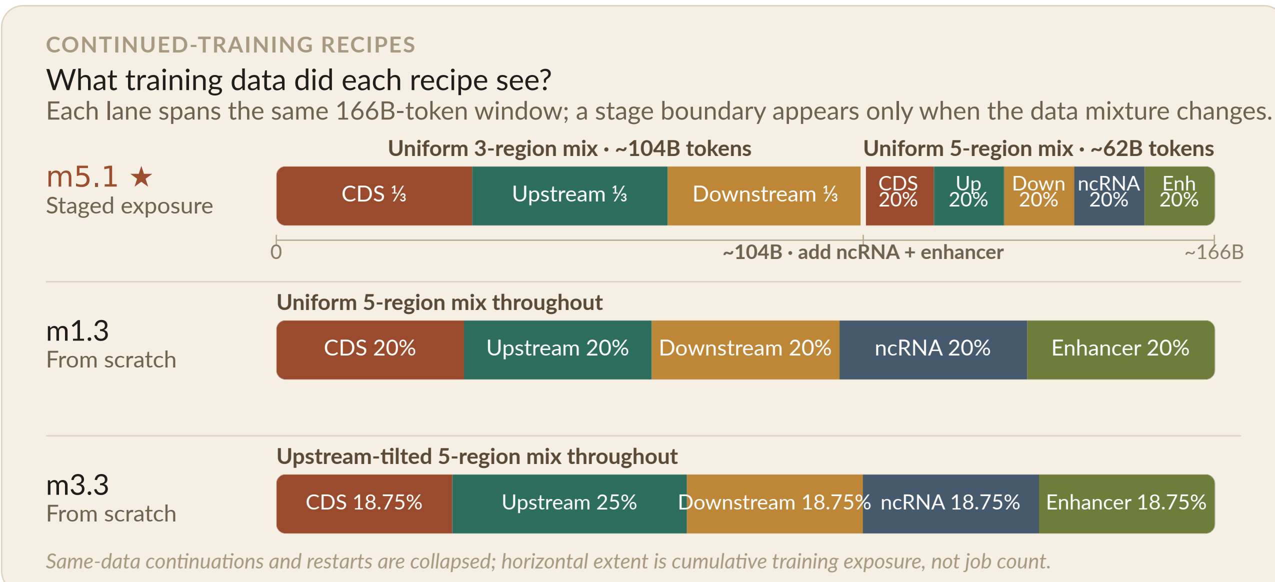

Figure 12 **Training-data exposure histories.** Three recipes are compared. m5.1 trains for approximately 104B tokens on a uniform three-region mixture before adding ncRNA and enhancer data for approximately 62B tokens. The de novo m1.3 and m3.3 controls keep fixed five-region mixtures over the same displayed token horizon; m3.3 gives upstream sequence 25% weight and each other region 18.75%.

The first evaluated m5.1 checkpoint after adding ncRNA and enhancer data improved on the corresponding ncRNA and distal subsets. At the final checkpoint, m5.1 had the highest macro-average AUPRC among the three lineages under both zero-shot LLR and the linear probe, although the linear-probe trajectories were noisier.[7] The controls retained stronger endpoints for some distal and ncRNA subsets. The endpoint pattern motivates a controlled test of exposure order but does not establish a benefit from staged training.

The corresponding linear-probe trajectories appear in Supplementary Figure S2.

[6] At the time, 1B had reached a good level of zero-shot performance under our then-current evaluation. We had not established it as the optimal model size. The later linear-probe results change this judgment most: they continued to improve with scale even where zero-shot Mendelian missense did not, making a larger model a more compelling choice in retrospect.

[7] The curves shown here use a separate probe trained within each variant subset. In a separate 255M analysis on one held-out chromosome, training one probe across all subsets improved the AUPRC point estimate on several data-starved subsets, including ncRNA and distal, while hurting stronger or more specialized subsets. This makes limited labeled data one plausible contributor to the noise, but that analysis did not directly test the checkpoint-to-checkpoint variability in these 1B lineage curves.

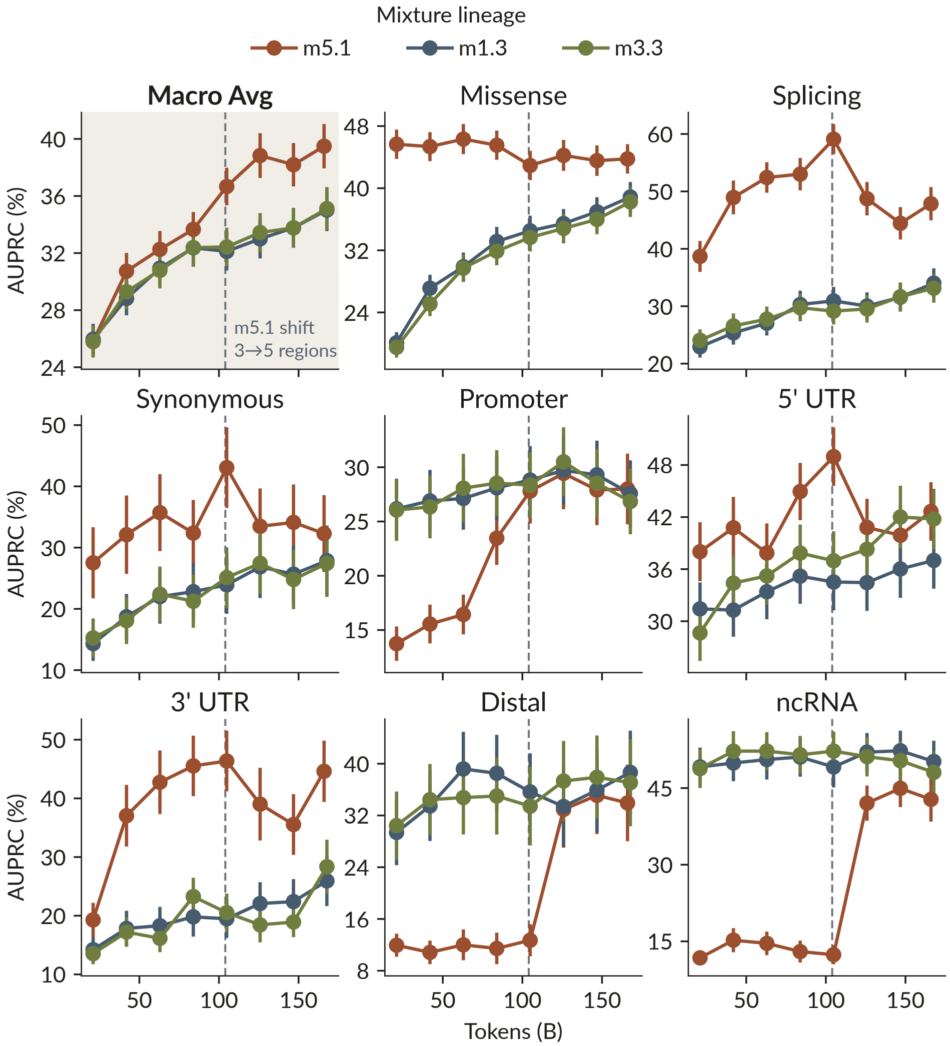

Figure 13 **Zero-shot mixture-lineage trajectories.** Mendelian AUPRC versus training tokens for three model-mixture lineages. Error bars denote SE.

2.6. Frozen comparison with external models

In the frozen zero-shot snapshot, the unweighted mean across the eight consequence subsets meeting the project-wide 30-positive minimum is 39.49% AUPRC for m5.1 and 38.24% for Evo 2 40B. The paired difference is 1.25 percentage points (bootstrap SE 1.63; 95% CI −1.93 to 4.46; two-sided bootstrap p = 0.440). This analysis contains 16,100 variants in 1,610 1:9 matched groups; the mature-miRNA subset, with four groups, is excluded from the macro-average by the same minimum applied throughout the evaluation pipeline. The confidence interval includes zero, so we describe m5.1 as statistically competitive with Evo 2 40B. Its point-estimate advantage is considerably larger under linear probing, for which this paired zero-shot comparison does not apply.

m5.1's three-stage lineage consumed 166.01B tokens and 1.135e21 recorded FLOPs. Evo 2 40B used 9.3T tokens and an estimated 2.25e24 FLOPs, giving a ratio of 1,982 that we report as approximately 1,980×. At their native context lengths on the same GH200, the optimized m5.1 configuration extrapolates to one million variants in about 41 minutes, compared with roughly 66 days for Evo 2 40B. The corresponding throughput ratio is approximately 2,330×.[8]

[8]This is an as-deployed workload comparison with forward- and reverse-complement passes and embeddings enabled. The m5.1 point uses delayed-FP8 throughput at 256 tokens and frozen BF16 AUPRC, while Evo 2 uses its original stack at 8,192 tokens. The full benchmark and quality-gate methodology is given in Section 4.10.

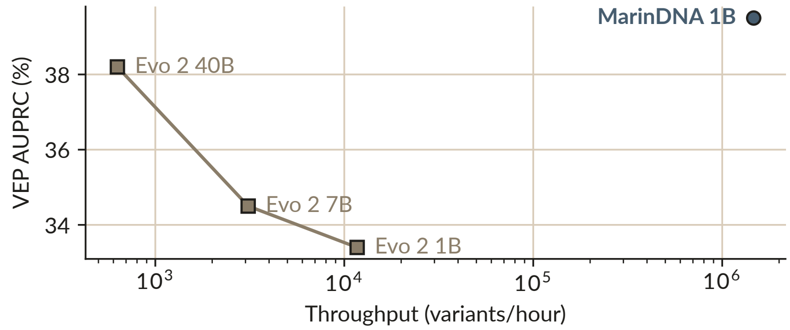

Figure 14 **Variant-effect performance versus as-deployed inference throughput.** Zero-shot Mendelian macro-average AUPRC versus variants scored per hour on one GH200. Each model is evaluated at its native context length; throughput includes forward- and reverse-complement scoring and embedding output. The MarinDNA point combines the frozen BF16 AUPRC with the optimized delayed-FP8 throughput; the FP8 configuration passed the paired zero-shot quality gate reported in Methods.

m5.1 outperforms Evo 2 40B on distal variants under both readouts. Evo 2 40B remains ahead on splicing under both readouts, on promoter and synonymous variants in the zero-shot evaluation, and on missense variants under linear probing.

Within the frozen leaderboard snapshot, no single MarinDNA model has the highest point estimate across all consequences. At least one other MarinDNA run has a higher point estimate than m5.1 in seven of eight displayed subsets, and several of those runs are region specialists. This pattern motivates further mixture experiments aimed at combining their complementary strengths.

In the broader frozen zero-shot comparison, m5.1's macro-average point estimate remains below AlphaGenome and GPN-Star.[9] AlphaGenome learns from functional-genomics supervision, while GPN-Star uses whole-genome alignments. These comparisons bound the alignment-free result without isolating the effects of architecture, training data, or supervision (research question #397).

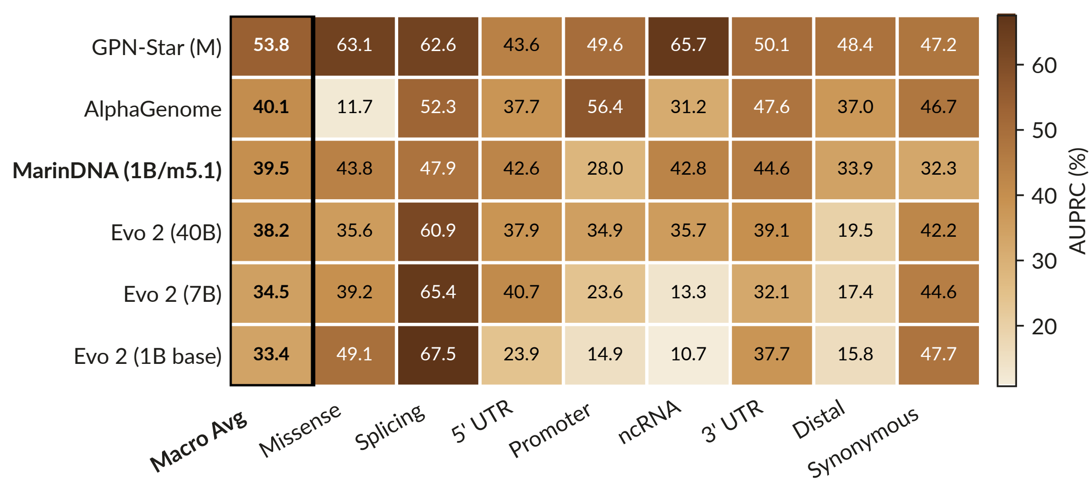

Figure 15 **Zero-shot Mendelian leaderboard.**

[9] Figure 15 uses LLR for MarinDNA and Evo 2, calibrated LLR (cLLR) for GPN-Star, and the maximum L2 REF/ALT prediction score for AlphaGenome.

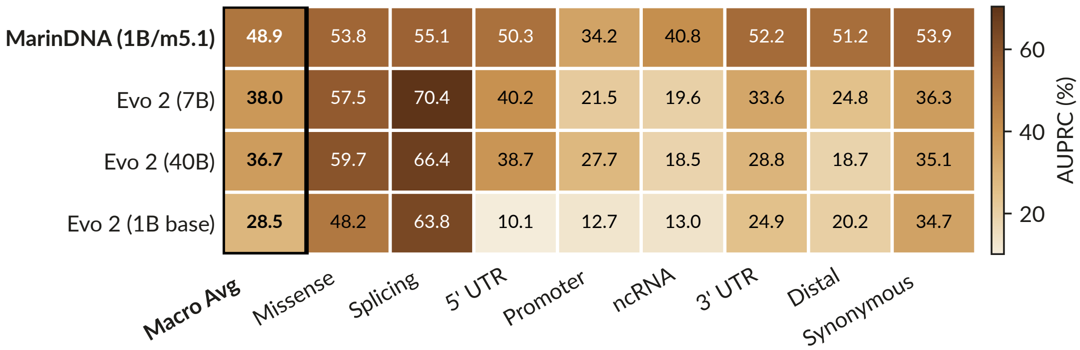

Figure 16 **Linear-probe Mendelian leaderboard.** AUPRC is a sample-size-weighted average across chromosomes and is not directly comparable with Figure 15. GPN-Star and AlphaGenome are absent because no compatible probe results are available here, not because of their performance.

3. Discussion

The final 1B MarinDNA model reached 39.49% macro AUPRC on the frozen Mendelian benchmark, compared with 38.24% for Evo 2 40B, while using 1,982-fold fewer reported training FLOPs. In native-context GH200 benchmarks, it scored variants at approximately 2,330 times the archived Evo 2 40B rate. The supporting experiments show how region sampling changed performance across consequence classes and how optimizer settings transferred across model size, batch size, and token horizon.

In the upstream/CDS experiment, proportional 10/90 pooling behaved similarly to CDS-only training, whereas equal sampling retained useful performance across both region types. The later m5.1 lineage introduced alignment-projected ncRNA and enhancer data and finished with the highest macro-average endpoint among the three frozen mixture lineages. These results support explicit control of region exposure. The mixture lineages were exploratory and do not establish that staged exposure is better than de novo five-region training.

The hyperparameter experiments provide complementary evidence that optimization settings can be transferred across model size, batch size, and token horizon in this regime. The transferred learning rate was the best observed tested value at each of three target scales, and the resulting parameter ladder trained stably through 4B parameters. The transfer rule nevertheless inherits assumptions from a text-model recipe, including its token-horizon exponent and model-geometry heuristic, and the present experiments do not establish that those choices are globally optimal for DNA.

On the frozen Mendelian evaluation, the final MarinDNA model's macro-AUPRC point estimate is 1.25 percentage points higher than Evo 2 40B, with a paired 95% bootstrap confidence interval from −1.93 to 4.46 percentage points. The interval includes zero, so the data support statistical competitiveness without demonstrating superiority. Point estimates differ by consequence class and readout: MarinDNA closes much of the distal-variant gap, while Evo 2 retains advantages on several coding and splicing subsets. GPN-Star and AlphaGenome have higher macro-average point estimates in the broader zero-shot comparison. They use alignment-derived information and functional-genomics supervision, respectively, so the comparison does not isolate architecture or training recipe.

The training-FLOP ratio uses MarinDNA's recorded lineage compute and Evo 2's published estimate, whose methodology explicitly omits mixed-precision and context-length adjustments. The throughput measurement compares steady-state scoring on the same GH200 at each model's native context length and includes forward- and reverse-complement passes plus embedding output; MarinDNA uses an optimized delayed-FP8 implementation, while the Evo 2 denominator is rounded at source. The throughput ratio measures the evaluated deployment configurations for this workload. It does not estimate architecture-only efficiency at matched context length or matched tokens processed. The FP8 configuration passed the zero-shot quality gate but was inconclusive as a drop-in source of embeddings for the existing BF16-trained probes, so the throughput result should not be generalized to probe compatibility.

The models use a short 255-base context chosen for functional-element evaluation, so these experiments do not test long-range regulatory interactions or genome-scale sequence generation. Human variant-effect data provide the strongest available evaluation, but performance in humans does not by itself establish transfer to non-model organisms, which is an important motivation for alignment-free models. The weighted validation loss is computed on fixed human sequences related to the training distribution and is useful for controlled comparisons. It is not an unbiased estimate of genomic generalization on a phylogenetically independent holdout. The zero-shot Mendelian missense regression with scale shows that likelihood-based readouts can diverge from information retained in frozen representations.

Linear-probe performance continued to improve through the 4B model, and the loss-scaling fit showed no clear saturation over the tested range. The experiments leave open whether larger models or further mixture and optimization changes would improve the frozen benchmarks.

4. Methods

4.1. Study design and frozen analysis snapshot

This study analyzes a frozen set of training runs, evaluation artifacts, and figures. All reported comparisons use fixed checkpoints and dataset revisions; the live leaderboard is a separate, evolving resource. The exact manuscript baseline and artifact revisions are recorded in Section 5 and Section 6.

4.2. Training data

Training examples were drawn from functional regions across multiple animal genomes. The initial datasets contained coding sequences (CDS), sequence upstream of annotated genes, and sequence downstream of CDS ends. The downstream category denotes the 256 bases immediately after an annotated CDS end rather than annotated 3′ UTR intervals. Later training mixtures added functional ncRNA regions and ENCODE V4 non-promoter candidate cis-regulatory elements, with human annotations projected to other species by sequence alignment. All genomic coordinates inside MarinDNA use 0-based, half-open intervals, with conversion to external coordinate conventions only at tool boundaries. The later recipe represents each example with 255 DNA bases and a beginning-of-sequence token, giving a 256-token model input. Soft-masked repetitive bases receive 1% of the standard training-loss weight. Training-data revisions and per-region token counts for every retained experiment will be enumerated in the supplementary provenance table.

4.3. Model and training objective

MarinDNA uses the Qwen3 decoder-only Transformer architecture with a causal next-token objective and nucleotide-level DNA tokenization. The architecture is intentionally conventional: the experiments vary training data, optimization, and parameter scale without introducing a genomics-specific sequence operator. The final m5.1 model has 1,120,772,224 parameters and a 255-base context. It was first trained for approximately 104B tokens on a uniform mixture of CDS, upstream, and downstream data, then continued for approximately 62B tokens after ncRNA and enhancer data were added to form a uniform five-region mixture. The frozen internal checkpoint is mix-v0.9-p1B-i24-exp135-m5.1-step-59158 at GCS path gs: The immutable public release is recorded in Section 5.

4.4. Hyperparameter calibration and transfer

A Bayesian reference sweep using Vizier trained approximately 25M-parameter models for 2.5B tokens with a 16k-token batch. The sweep tuned initialization scale, AdamH and Adam peak learning rates, β_1, β_2, ε, gradient clipping, and z-loss against the weighted validation-loss statistic described below. A fixed geometry rule maps hidden width to the number of layers, MLP width, attention heads, and key-value heads. For hidden width D, it sets `layers = round(D/(55 + 4·log₂D))`, `MLP width = 4D`, and both attention-head and key-value-head counts to `D/128`; sweep widths are selected manually. The rule is inherited from Marin's text-model scaling heuristic and was not fit on DNA; see the commit-pinned geometry rule and model builder. A second fixed rule transfers the reference optimizer settings to a target batch size and token horizon. Relative to reference batch size B_0 and horizon T_0, the AdamH learning rate is multiplied by $\sqrt{(B/B_0)}(T_0/T)$^0.3. The Adam learning rate is multiplied by $\sqrt{(r/r_0)}$, ε by $\sqrt{(r_0/r)}$, and β_2 is exponentiated by B/B_0 subject to configured bounds, where $r/r_0 = (B\ T_0)/(B_0\ T)$. The 0.3 token-horizon exponent is inherited from Marin's text recipe and was not fit on the DNA sweep. Complete(d) proposes 0.5, while a later Marin text sweep estimated ~0.28; see the commit-pinned implementation. Initialization scale, β_1, gradient clipping, and z-loss are reused without transformation. Both learning rates use linear warmup over the first 10% of steps, remain at their peak through 80% of training, and decay linearly to zero over the final 20%. See the experiment configuration and scheduler implementation. Target-scale validation used 255M-, 476M-, and 1B-parameter models with four times the token horizon and one quarter of the reference batch size. "Best observed" refers to the lowest final validation loss among the tested target-scale settings and does not imply a global optimum.

4.5. Parameter-scaling experiment and validation loss

The parameter ladder contains models of approximately 46M, 76M, 128M, 255M, 476M, 1B, 2B, and 4B parameters. Each model was trained for approximately 84.77B tokens with the transferred optimization recipe. The final validation losses were fit with a Kaplan-style power law as a function of parameter count. During training, lowercase marks repetitive bases; in validation, it marks non-conserved bases. Both receive 1% of the standard token weight. The validation statistic is computed on fixed human sequences related to the training distribution and is used for controlled comparisons and model selection. It is not interpreted as an unbiased estimate on a phylogenetically independent held-out genome set. We did not independently validate the lowercase weighting as a model-selection objective. Because the statistic informed the reference sweep and the choice of the 1B model, another validation objective could select different settings. The biological conclusions rely on downstream VEP evaluation.

4.6. Variant-effect datasets

The Mendelian benchmark compares pathogenic variants with non-rare gnomAD controls matched within consequence class. Relative to the TraitGym design, the minimum control allele frequency is 0.1%, the consequence set includes missense and splicing variants, and matching includes potential confounders such as transcription-start-site distance and exon distance where applicable. The frozen benchmark revision is 4aed58e50c5dea0b878a665007af2ef9e5108e9f. The saturation-genome-editing benchmark is the v3 labeled build at 225d3d1ea32a4af547891b13c33b5e92a5aae849. It retains variants labeled abnormal or normal by ClinGen/ExCALIBR-calibrated assay thresholds and groups them into missense and splicing subsets. Variant contexts are extracted from the Ensembl release 115 GRCh38 soft-masked primary assembly. The evaluation pipeline uses the development split for model iteration; the chromosome-disjoint test split remains reserved for the final locked evaluation.

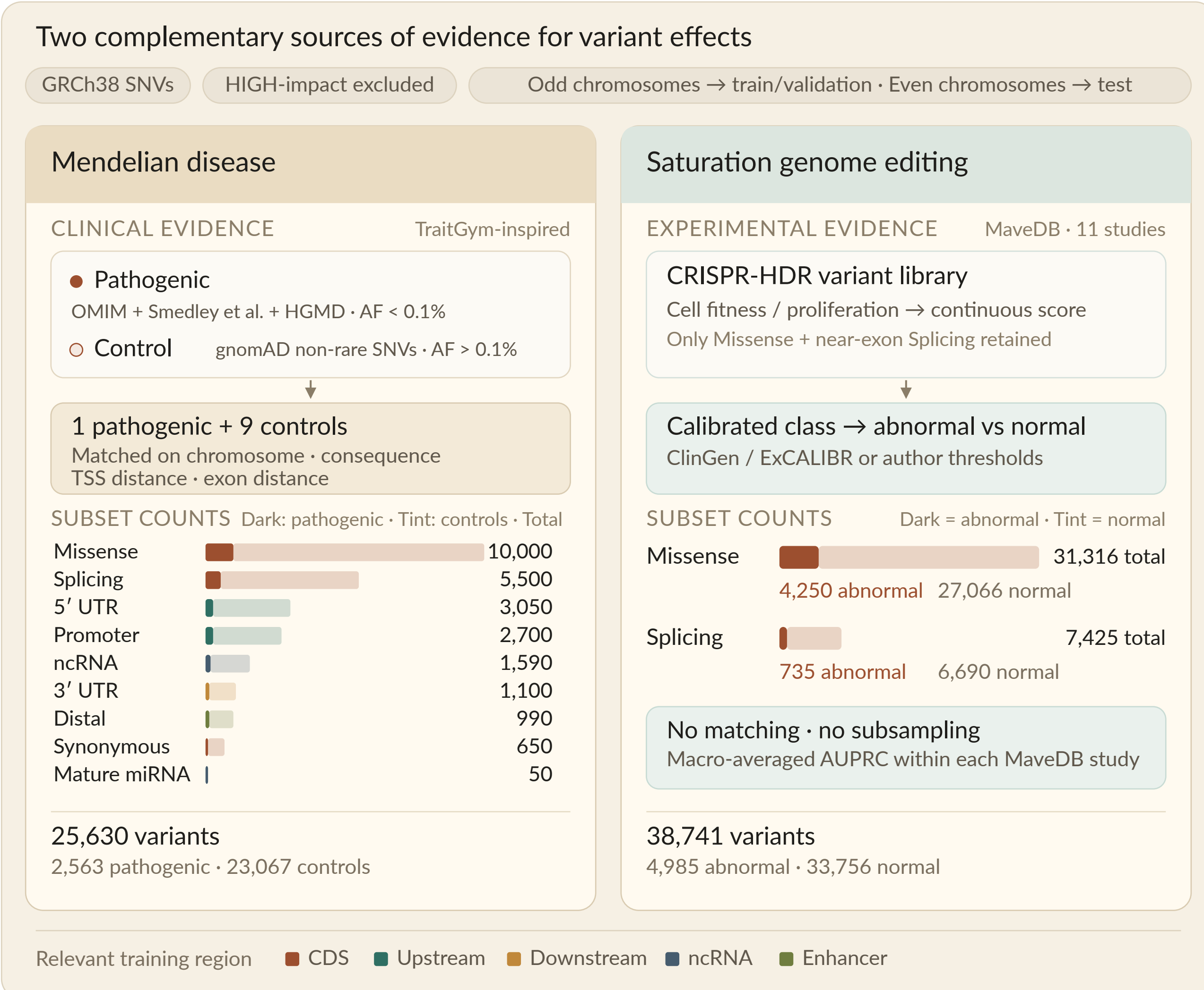

Figure 17 **Variant-effect prediction benchmarks.** Counts denote the frozen analysis snapshot described in the text.

The two benchmarks and their analysis subsets are summarized in Figure 17.

4.7. Zero-shot likelihood scoring

For each allele, the causal model scores the variant-centered sequence on the forward strand and its reverse complement. The raw log-likelihood ratio is defined as log P(ALT) − log P(REF). For Mendelian and SGE evaluation, deleteriousness is oriented as the negative of the mean forward- and reverse-complement log-likelihood ratio, so larger scores indicate a larger likelihood penalty for the alternate allele. Forward and reverse-complement scores are retained separately, but their mean is the default MarinDNA and Evo 2 score used in the headline comparison. GPN-Star is represented by its calibrated likelihood-ratio score, whereas AlphaGenome is represented by its maximum L2 reference/alternate prediction score; these scores are not interchangeable measurements of model likelihood.

4.8. Frozen-embedding linear probes

The same forward and reverse-complement passes used for likelihood scoring also produce last-layer hidden states for the reference and alternate alleles. Hidden states are mean-pooled across the entire DNA window with special tokens excluded, accumulated and strand-averaged in float32, and stored in float16. For directional Mendelian and SGE labels, the probe feature concatenates the reference embedding with the alternate-minus-reference embedding after upcasting to float32. A separate StandardScaler plus L2-regularized logistic-regression pipeline is fit within each consequence subset. Predictions use leave-one-chromosome-out cross-validation, with the regularization strength retuned inside each outer fold by chromosome-grouped cross-validation over a fixed 17-point grid from 1e-12 to 1e4. Probe and zero-shot metrics are calculated on identical variant rows.

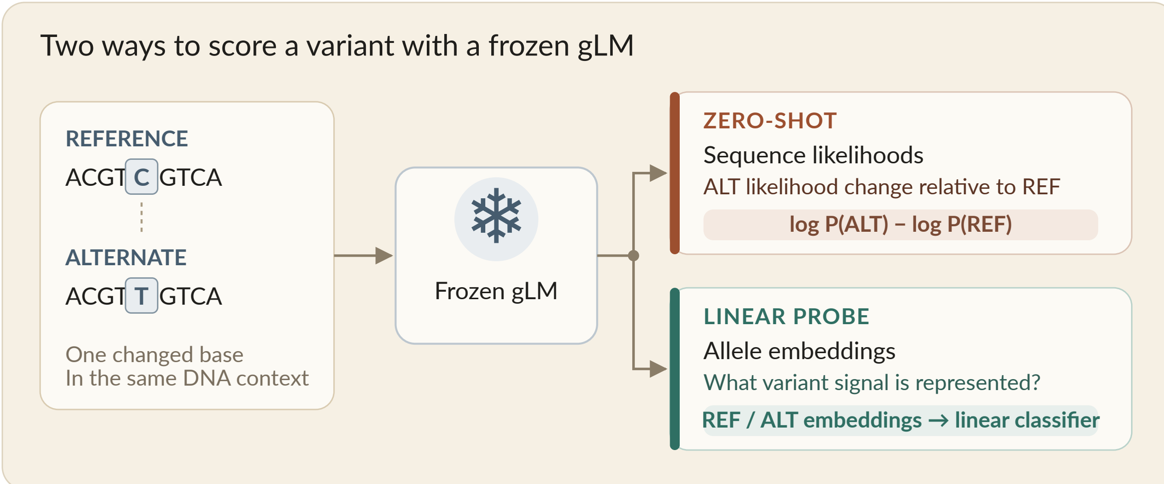

Figure 18 **Variant-effect prediction scoring approaches.** The zero-shot likelihood score and frozen-embedding linear probe use the same reference and alternate sequence inputs.

Figure 18 contrasts the two readouts applied to every frozen checkpoint.

4.9. Metrics and uncertainty

Performance is summarized by area under the precision–recall curve (AUPRC). For matched Mendelian zero-shot analyses, AUPRC is computed within consequence subset and uncertainty is estimated by resampling matched groups with replacement, thereby preserving the one-to-many matched structure. For probe analyses, AUPRC is first calculated per chromosome and then weighted by the number of contributing variants; uncertainty is estimated by a chromosome-cluster bootstrap. SGE AUPRC is computed separately within each MaveDB accession and consequence subset before macro-averaging because scores are not comparable across studies. Unless a caption states otherwise, error bars denote ±1 bootstrap standard error. The headline comparison uses the eight consequence subsets with at least 30 positive matched groups, the project-wide minimum for inclusion in the Mendelian macro-average. It contains 16,100 variants in 1,610 groups, each comprising one positive and nine matched controls; the ninth subset, mature miRNA, has four groups and is excluded from the macro-average. The MarinDNA and Evo 2 score artifacts contain the same 16,140 uniquely keyed variants and agree exactly on label, subset, and match-group assignment. For the paired comparison, 10,000 bootstrap iterations are generated with random seed 0. Within each consequence subset, matched groups are sampled with replacement; the same sampled rows are used for both models, AUPRC is recomputed for each, and the eight within-subset differences are averaged with equal weight. The comparison reports the observed macro-AUPRC difference, its bootstrap standard error, the 2.5th and 97.5th percentiles of the paired difference distribution, and a two-sided bootstrap p-value calculated as twice the smaller empirical probability that the paired difference is at or below zero or at or above zero.

4.10. Training compute and inference throughput

MarinDNA training compute is reconstructed from the recorded W&B `throughput/total_gflops` fields at the boundaries of all three segments inherited by m5.1. The lineage contains 166,010,552,320 tokens and 1.13498e21 FLOPs. Evo 2 40B reports 9.3T tokens and an estimated 2.25e24 FLOPs; the source table states that the estimate does not account for mixed precision or pretraining context length (Brixi et al., 2026). The ratio of the reported values is 1,982, rounded to approximately 1,980-fold.

Inference throughput was measured on one 96-GB NVIDIA GH200 at USD 2.29 per hour with forward- and reverse-complement passes and pooled reference/alternate embedding output enabled. The optimized m5.1 measurement used PyTorch 2.10.0+cu128, Transformer Engine 2.13 delayed E4M3 FP8, fused MLP and QKV projections, `torch.compile` in default mode, batch size 128, four data-loader workers, and three synchronized repetitions on 5,800 pre-materialized variants. It achieved 1,467,776 variants per hour, equivalent to 2,452.69 seconds per million variants. The plotted m5.1 AUPRC remains the frozen BF16 result, whereas its throughput coordinate uses this optimized FP8 configuration. The promoted FP8 model passed a separate 10,000-draw paired zero-shot quality gate: BF16 minus FP8 macro AUPRC was 0.000355 (95% interval −0.004271 to 0.005190). A frozen-BF16-probe compatibility check was inconclusive, so the optimized rate supports zero-shot scoring and the stated embedding-output contract but not an unconditional drop-in claim for existing BF16-trained probe classifiers. The Evo 2 40B measurement used the NVIDIA PyTorch 25.04 container plus the Evo 2 inference package, batch size 1, and a late converged batch rate after model loading, warm-up, and tokenization; its archived rate is rounded to 630 variants per hour. Both measurements are steady-state and exclude sequence materialization. The resulting ratio is approximately 2,330-fold, but its precision is limited by the rounded Evo 2 rate. The models use their native context lengths, 256 tokens for m5.1 and 8,192 tokens for Evo 2 40B, so this is an as-deployed workload measurement rather than a same-context, same-batch, or per-token architecture benchmark. The full benchmark record is available in issue #354.

5. Data, Code, Models, and Live Resources

The MarinDNA repository contains the data pipelines, evaluation implementation, manuscript source, and tracked investigation scripts. The mechanical manuscript and frozen figure baseline is available at commit 3b608d39b41c2330636ec647dbb25d26b0895187. The final preprint revision will identify commit-pinned training, dataset-construction, evaluation, and plotting code for each retained result.

The frozen Mendelian benchmark is marin-dna/evals_mendelian_traits at revision 4aed58e. The frozen saturation-genome-editing benchmark is marin-dna/evals_sge at revision 225d3d1. The row-aligned MarinDNA–Evo 2 headline comparison is reproduced by a commit-pinned audit script, which names both frozen score artifacts and asserts their variant, label, subset, and match-group identity before bootstrapping. The reported compute and throughput ratios are reconstructed by a separate commit-pinned efficiency audit. The released m5.1 checkpoint is marin-dna/marin-dna-exp135-m5.1 at revision c0676b2. Training datasets, evaluation datasets, and the released model are grouped in the MarinDNA Hugging Face collection. The exact training-dataset revisions will be repeated in the supplementary provenance table before the preprint is released.

The MarinDNA leaderboard is an evolving project resource. Its contents may change as models and baselines are added and must not be treated as the frozen comparison reported in this manuscript. The committed manuscript figures and their recorded input revisions define the archival result snapshot.

Optional interactive resources include:

- Interactive sequence explorer.
- Model inference and BRCA1 variant-effect prediction notebook.

6. Open Development and Provenance

MarinDNA was developed through public, issue-tracked experiments in which hypotheses, configurations, intermediate results, and negative findings were recorded as the work progressed. The issue history provides development provenance but is not required scientific exposition: the manuscript states the methods, evidence, and limitations needed to interpret its claims. Decisions and substantial editorial changes for this frozen preprint are tracked in issue #449. The mechanical manuscript baseline and its 20 SVG assets were fixed at MarinDNA commit 3b608d39b41c2330636ec647dbb25d26b0895187, whose content and figures were converted from project commit d8c4803cbbbffafb24890cd0c75134d78368d55c. No new training run or evaluation dataset was introduced during that conversion.

Key result families retain the following development records, which will be expanded with exact comments, code commits, checkpoints, dataset revisions, and figure inputs in the supplementary provenance table:

- Data specialists and mixture balance: #21, #27, and #13.
- Manual scaling failure and the need for transfer: #57.
- Validation-loss design and interpretation: #8.
- Frozen-embedding probe development: #369 and the evaluation-pipeline records it references.
- Frozen MarinDNA–Evo 2 comparison: the refreshed Evo 2 score record and the commit-pinned paired analysis.

- Training compute and throughput: #135, #354, #430, and the commit-pinned arithmetic audit.

Issue links are provided to make the branching research history inspectable; scholarly claims about prior work use bibliographic citations, and claims about MarinDNA are supported by the manuscript's figures, Methods, and frozen artifacts.

7. Acknowledgements

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8. Author Contributions

TODO: Complete after the author list and order are agreed.

9. Competing Interests

TODO: Obtain and record declarations from all authors before public posting.

10. Funding and Compute Resources

Training used compute resources provided by the Google TPU Research Cloud program. Additional funding and resource disclosures will be completed before public posting.

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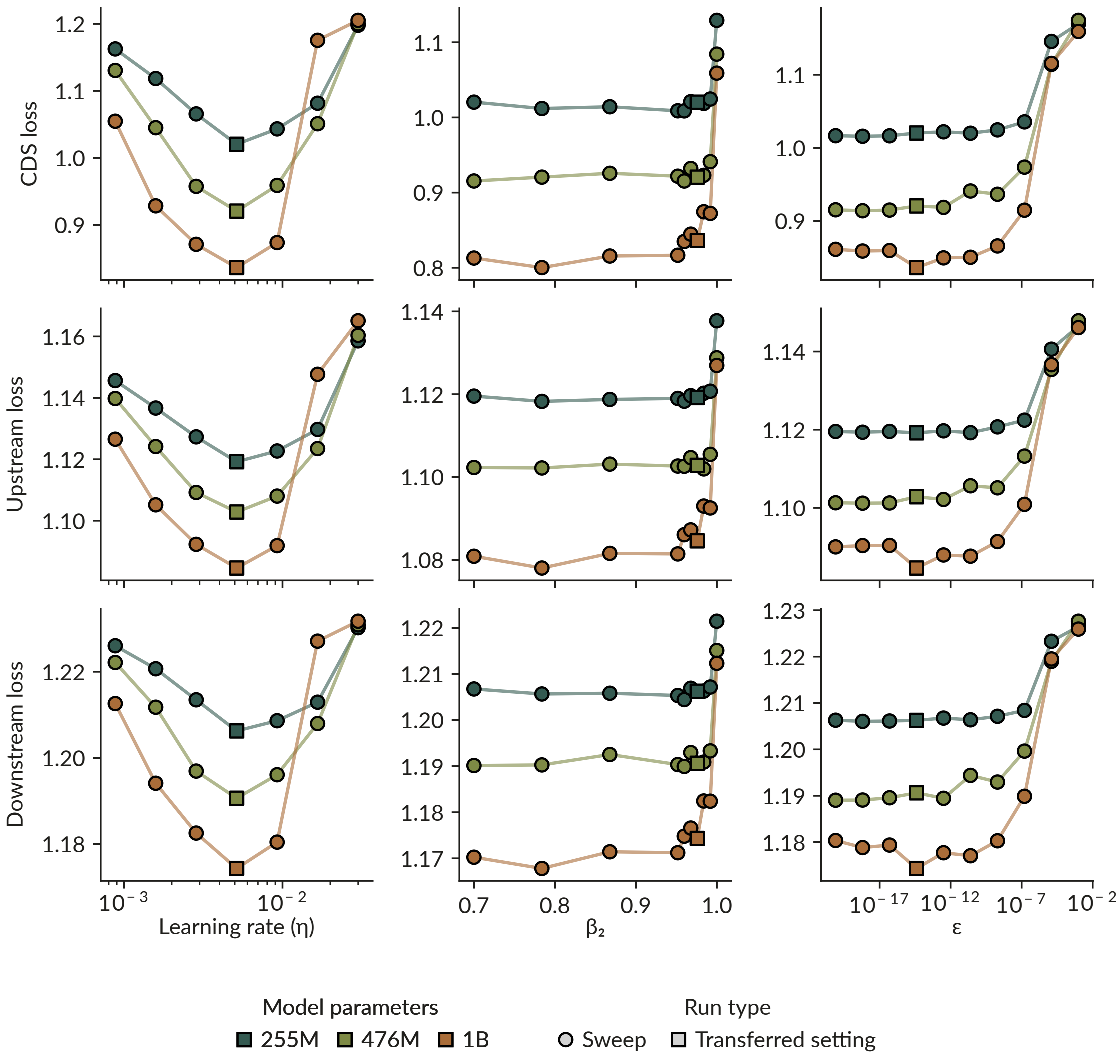
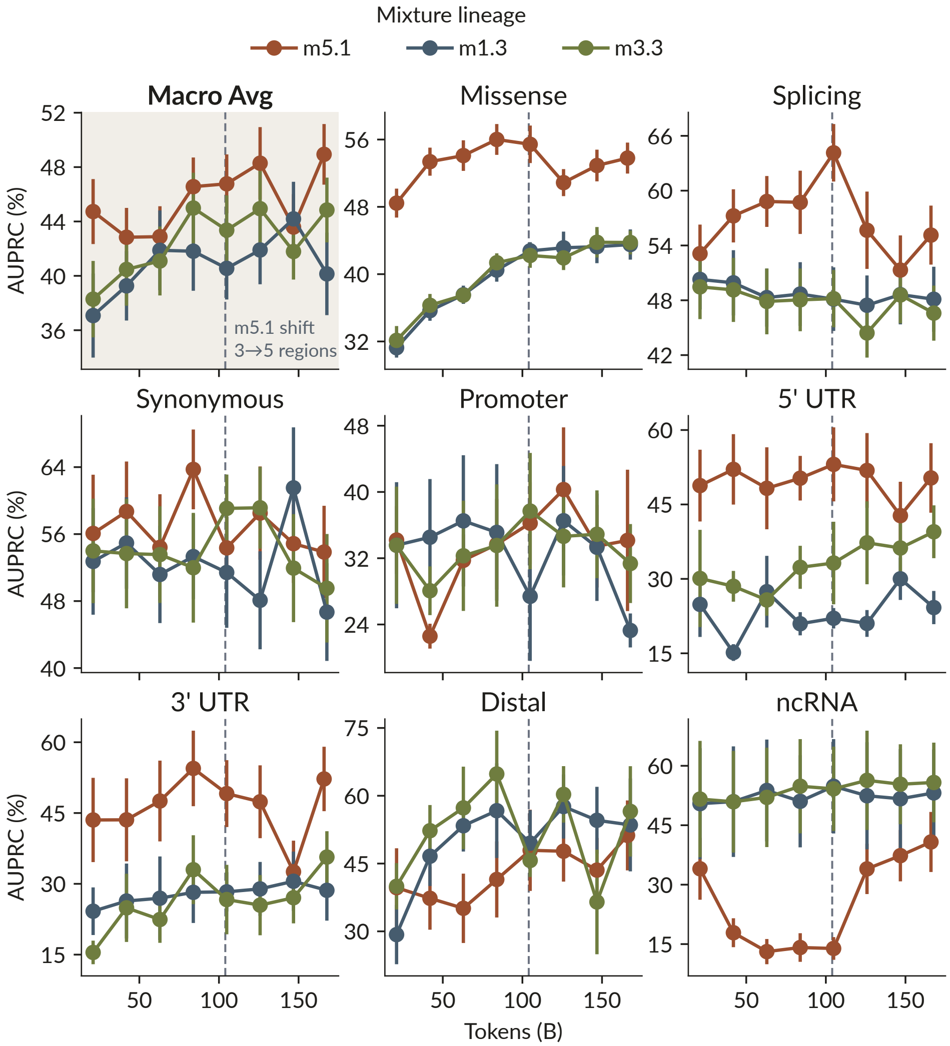
Supplementary Information

The supplementary figures report region-stratified hyperparameter-transfer results and the linear-probe trajectories corresponding to the zero-shot mixture analysis in Results.



Supplementary Figure S1 **Region-specific hyperparameter transfer.** Validation loss as a function of learning rate, β_2, and ε, evaluated separately for CDS, upstream, and downstream regions.



Supplementary Figure S2 **Linear-probe mixture-lineage trajectories.** Linear-probe Mendelian AUPRC (chromosome-averaged) versus training tokens for three model-mixture lineages. Error bars denote SE.